

Ingredient Evidence — Physician Review

Cited efficacy studies for the ingredient library, prepared for medical sign-off before publication.

21	58	2	0	19
ingredients	cited studies	established	emerging	limited

For Dr. Nguyen. Every citation below was drawn from PubMed, PMC, NCBI, or Cochrane and independently fact-checked to confirm it is a real study whose finding is not overstated. The verification pass removed citations that failed (e.g. a vitamin D study whose finding was reversed, and a zinc trial under a journal Expression of Concern). Wording is deliberately conservative and CPSO-safe: most IV use in healthy, replete people is rated *limited*, which reflects the literature honestly. Please confirm or revise each ingredient, and sign at the end. Nothing publishes as a treatment claim without your approval.

Established — essential role / clinical use is well supported. **Emerging** — promising, human evidence still developing.

Limited — little/no direct evidence for IV use in healthy people.

PUBLISHES AS: Strong evidence for correcting deficiency (oral or topical); IV use in healthy, replete people is not established.

1. Oral vitamin A reduces night blindness and xerophthalmia in young children

A large systematic review and meta-analysis of trials in children under 5, mostly in settings where deficiency is common, found that oral vitamin A supplementation was associated with less night blindness (RR 0.32, 95% CI 0.21-0.50) and less xerophthalmia (RR 0.31, 95% CI 0.22-0.45), along with lower all-cause mortality (about 24%, RR 0.76) and less illness. The evidence is for oral dosing to prevent or correct deficiency, not intravenous vitamin A in well-nourished people.

APPLICABILITY: Applies mainly to oral vitamin A in deficient children; it does not address IV vitamin A for vision in already-replete healthy adults, so it may support correcting deficiency but says little about IV wellness use.

SOURCE: Mayo-Wilson et al., BMJ systematic review and meta-analysis, 2011 (BMJ 2011;343:d5094) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC3162042/>

2. Topical retinol may improve the appearance of photodamaged skin

A double-blind, vehicle-controlled trial reported that one year of topical stabilized retinol improved the appearance of photodamaged facial skin in adults. This is a skin-applied (topical) application, not IV or oral systemic dosing.

APPLICABILITY: The skin benefit here comes from retinol applied directly to the skin; it does not demonstrate that IV vitamin A supports skin in healthy people.

SOURCE: Randhawa et al., J Drugs Dermatol 2015 (Cochrane CENTRAL record CN-01087680) · <https://www.cochranelibrary.com/central/doi/10.1002/central/CN-01087680/full>

FACT-CHECK Both citations verified as real and accurately represented; both retained. Study 1 is genuinely the Mayo-Wilson BMJ 2011 review (PMC3162042); confirmed the specific vision figures (night blindness RR 0.32, xerophthalmia RR 0.31) and oral route, and named the authors/citation for verifiability. Study 2 is genuinely the Randhawa 2015 topical retinol trial (Cochrane CENTRAL CN-01087680); replaced the long tracking-parameter URL with the clean canonical Cochrane record URL, added the primary journal/author attribution, and softened the finding title to 'may improve.' Both summaries already correctly flag topical/oral vs IV, so wording is CPSO-safe. evidenceLevel kept at 'limited' because, while the underlying evidence is solid for oral use in deficient children and topical use on skin, neither supports IV vitamin A in healthy, replete people — which is the relevant clinical claim. NCBI and Cochrane returned automated-access 403s, so content was confirmed via corroborating search results rather than direct fetch.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Clear roles in immunity and collagen, but IV benefit in healthy people is limited and still being studied.

1. **IV vitamin C may improve some neutrophil immune functions**

Systematic review of 16 randomized controlled trials examining vitamin C and neutrophil function, largely in hospitalized, exercising, or critically ill patients. Overall about 44% of the RCTs showed an effect on some neutrophil function (e.g., chemotaxis), and two of three chemotaxis trials showed improvement after intravenous vitamin C. Results were heterogeneous across differing cohorts and dosing regimens.

APPLICABILITY: Populations were patients (hospitalized, post-exercise, critically ill) rather than healthy wellness clients, and effects were inconsistent (about 44% of trials), so this may support but does not prove an immune benefit from a wellness drip.

SOURCE: Nutrients, 2019 systematic review of RCTs (PMC6770220) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC6770220/>

2. **Vitamin C may support collagen synthesis after injury**

Systematic review of 10 studies (animal and human, mostly oral supplementation) reporting that vitamin C may increase collagen synthesis, support bone/tendon/ligament healing, and reduce oxidative stress markers after musculoskeletal injury.

APPLICABILITY: Evidence is largely oral supplementation in injured or animal models, not intravenous dosing in healthy people; the authors note clinical evidence is limited and more human trials are needed.

SOURCE: Orthopaedic Journal of Sports Medicine, 2018 systematic review (PMID 30386805) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC6204628/>

3. **Vitamin C may support tissue and wound healing**

Systematic review of 18 studies concluding that vitamin C supplementation may improve healing outcomes in certain conditions, predominantly pressure ulcers, consistent with vitamin C's role in collagen formation and tissue repair.

APPLICABILITY: Benefits were seen mainly in patients with impaired healing or likely deficiency, often with small samples, combined nutritional treatments, and mostly oral routes; this does not establish a collagen or healing benefit from IV vitamin C in healthy, replete individuals.

SOURCE: Antioxidants (Basel), 2022 systematic review on vitamin C in tissue healing (PMC9405326) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC9405326/>

FACT-CHECK All 3 citations verified as real studies that load and are accurately represented; none removed. Corrections: (1) Study 1's journal was wrong — it is Nutrients (MDPI, 2019), not "Antioxidants"; source fixed and PMC ID added. Confirmed 16 RCTs and the ~44% figure are accurate, and IV route was used in the chemotaxis trials. (2) Study 2 source refined to name the journal (Orthopaedic Journal of Sports Medicine, 2018); PMID and 10-study/animal+human/oral details confirmed accurate. (3) Study 3 confirmed (Antioxidants 2022, 18 studies, predominantly pressure ulcers); softened the fibroblast-proliferation phrasing to avoid overstating a mechanistic claim, and kept it CPSO-safe. evidenceLevel remains "limited" — justified by heterogeneous results, predominantly oral/animal routes, and patient (not healthy-client) populations, with sparse IV-specific evidence. All summaries use "may support" language with no guarantees.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Well-defined for bone and immune health when you run low; the benefit is oral and deficiency-driven, not IV.

1. May modestly reduce acute respiratory infections in deficient people

Systematic review and meta-analysis of aggregate data from randomised controlled trials (oral vitamin D) reported a modest protective effect against acute respiratory infections. Benefit was concentrated in people who were vitamin D deficient at baseline and received daily or weekly (not bolus) dosing; overall effects were small.

APPLICABILITY: All trials used oral dosing; any benefit was concentrated in deficient individuals on regular daily/weekly dosing and was modest and mixed across analyses. There is no evidence for IV vitamin D or for benefit in healthy, vitamin-D-replete adults.

SOURCE: Lancet Diabetes & Endocrinology, 2021 (systematic review and meta-analysis of aggregate RCT data; PMC) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7709175/>

2. Calcium plus vitamin D may support fracture risk reduction; vitamin D alone is limited

Updated meta-analysis of RCTs from the National Osteoporosis Foundation found that calcium plus vitamin D (oral) supplementation was associated with a reduced risk of total fractures. This meta-analysis specifically assessed the calcium-plus-vitamin-D combination; evidence for vitamin D alone on fractures and bone density is more limited and inconsistent.

APPLICABILITY: The fracture benefit was observed for oral calcium combined with vitamin D, generally in older/deficient populations. Vitamin D alone and IV administration are not supported for bone health in healthy people.

SOURCE: Osteoporosis International, 2016 (National Osteoporosis Foundation updated meta-analysis; PMC) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC4715837/>

FACT-CHECK Removed the muscle-strength citation (PMID 34405916). It is a real meta-analysis (Bislev et al., 54 trials, 8,747 participants) but was doubly misrepresented: (1) misattributed to Calcified Tissue International when it was published in the Journal of Bone and Mineral Research (2021), and (2) its finding was reversed — the study concluded vitamin D does NOT benefit muscle strength and may have a small harmful effect, the opposite of the claimed "small positive effect." Kept the respiratory-infection study (PMC7709175, correctly the Jolliffe 2021 Lancet Diab Endocrinol aggregate-data meta-analysis) and the fracture study (PMC4715837, Weaver 2016 NOF meta-analysis), both of which loaded as real studies and matched their claims; summaries were softened to CPSO-safe "may support" language and clarified (respiratory benefit is modest/deficiency-limited/oral; fracture benefit is specifically for calcium+vitamin D). Downgraded evidenceLevel from "emerging" to "limited": the only study asserting a direct positive benefit was removed, and the two survivors show only modest, deficiency-conditional, mixed oral effects with no support for IV use or benefit in healthy replete adults.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: A recognised antioxidant, but human trials for skin protection are largely negative or topical, not IV.

1. Oral vitamin E did not meaningfully reduce UV oxidative stress

In healthy human volunteers, oral vitamin E significantly reduced one marker of skin lipid oxidation (malondialdehyde), but neither vitamin E nor beta-carotene affected other measures of UV-induced oxidative stress or skin sensitivity to UV, so the authors concluded there was no meaningful photoprotection from supplementation.

APPLICABILITY: Oral route in healthy people, but largely negative for skin protection; does not support an IV skin-protective claim.

SOURCE: Am J Clin Nutr / PubMed, 2004 · <https://pubmed.ncbi.nlm.nih.gov/15531675/>

2. Topical vitamin C plus E with ferulic acid reduced UV skin damage

A stabilized topical solution of vitamins C and E with ferulic acid provided measurable protection of human skin against UV-induced damage, including reduced thymine dimers. This was a topical formulation applied to the skin, not systemic or IV vitamin E.

APPLICABILITY: Topical application only; results do not transfer to an intravenous drip, which delivers vitamin E systemically rather than to the skin surface.

SOURCE: J Am Acad Dermatol / PubMed, 2008 · <https://pubmed.ncbi.nlm.nih.gov/18603326/>

3. Narrative review of vitamin E roles in skin

A dermatology review summarizes vitamin E's antioxidant and free-radical-scavenging properties in skin and notes that, despite experimental photoprotective and antitumorigenic activity, controlled clinical evidence for defined dosing and indications is sparse.

APPLICABILITY: Narrative review, not an RCT; describes mechanisms but does not establish that IV vitamin E protects skin cells in healthy people.

SOURCE: Indian Dermatol Online J review (PMC), 2016 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC4976416/>

FACT-CHECK All 3 citations verified as real via NCBI-indexed search (the pubmed/eutils hosts are blocked by the egress proxy, so abstracts were confirmed through WebSearch, not by routing around the denial). Kept all 3. Corrected study 1's journal, which was misattributed: it is Am J Clin Nutr 2004, not J Invest Dermatol. Tightened study 1's summary to attribute the malondialdehyde reduction to vitamin E specifically and to note UV-sensitivity was unaffected. Minor accuracy edit to study 2 (added that it reduced thymine dimers; reaffirmed topical route). No overstatements found; all three are honestly framed as not supporting an IV vitamin-E skin claim. evidenceLevel kept at "limited," which is appropriate given the negative oral RCT, a topical-only study, and a narrative review.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Essential cofactors in energy metabolism; trials are small and oral, with no IV evidence in replete people.

1. Oral B-complex reduced exercise fatigue markers in healthy adults

In a randomized, double-blind crossover trial of 32 healthy non-athletes (aged 20-30), 28 days of an ORAL vitamin B complex supplement was associated with longer running time to exhaustion and lower blood lactate and ammonia during and after exercise. This is consistent with B vitamins' established cofactor role in energy metabolism and does not demonstrate an energy 'boost' in already-replete people.

APPLICABILITY: Directly on B-complex and exercise fatigue, but the route was ORAL (a commercial oral formulation), not IV, and subjects were healthy young adults; it does not show that an IV drip works in non-deficient people or improves subjective energy.

SOURCE: Int J Med Sci, 2023; randomized double-blind crossover trial · <https://pubmed.ncbi.nlm.nih.gov/37786445/>

2. Oral high-dose B multivitamin linked to brain metabolite and oxidative-stress markers

A randomized, double-blind, placebo-controlled trial in 32 healthy adults (aged 30-65) found that high-dose ORAL B-group vitamins raised vitamin B6 and B12 and lowered homocysteine, and identified correlations between B-vitamin levels and brain metabolites measured by MRS. The authors described this as preliminary evidence that B vitamins may support oxidative metabolism; the key brain metabolite (NAA) increase was not statistically significant, and no change in felt energy was measured.

APPLICABILITY: Suggests a plausible metabolic mechanism, but dosing was ORAL and the brain findings were largely correlational and preliminary; provides no evidence specific to intravenous administration or to subjective energy in healthy individuals.

SOURCE: Nutrients, 2018; randomized, double-blind, placebo-controlled trial · <https://pmc.ncbi.nlm.nih.gov/articles/PMC6316433/>

FACT-CHECK Both citations verified as real, correctly-attributed studies. NCBI hosts were blocked by the egress proxy (403), so I confirmed each via the publisher (Int J Med Sci / medsci.org) and PMC/Nutrients metadata through web search rather than direct fetch. Study 1 (PMID 37786445, Int J Med Sci 2023) matched its citation; I corrected the source year/journal, noted it was a crossover using a commercial oral product, and dropped the unverified specific 'B1, B2, B6, B12' composition claim. Study 2 (PMC6316433, Nutrients 2018) was overstated: its finding title said the supplement 'altered brain oxidative metabolism,' but the actual brain metabolite (NAA) increase was NOT statistically significant ($p > 0.05$) and the effects were correlational/preliminary — I rewrote the finding and summary to reflect this and kept CPSO-safe 'may support' language. evidenceLevel remains 'limited' and is accurate: both trials are small ($n=32$), ORAL, and none tests IV administration in non-deficient people.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Vital for nerve and blood health when low; supplementation shows little benefit in people who are not deficient.

1. B12 supplementation showed no benefit for cognition/mood without deficiency

Systematic review, meta-analysis and meta-regression of 16 RCTs (6,276 participants) of vitamin B12, largely oral and often with folic acid/B6, in people without overt B12 deficiency or advanced neurological disease. It found no effect on cognitive function or depressive symptoms, and only one included trial reported idiopathic fatigue, so fatigue could not be pooled. The authors concluded supplementation is likely ineffective for these outcomes in non-deficient people.

APPLICABILITY: Tempers the energy/mood claim: in non-deficient people (as most IV wellness clients are) B12 may not support cognition or mood, fatigue evidence was too sparse to assess, and the trials used oral, not IV, dosing.

SOURCE: Nutrients, 2021 (systematic review, meta-analysis and meta-regression) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8000524/>

2. Oral methylcobalamin improved several neuropathy measures in type 2 diabetes

One-year double-blind, placebo-controlled RCT in 90 patients with type 2 diabetes (on long-term metformin, B12 below 400 pmol/L) and peripheral and autonomic neuropathy, given oral methylcobalamin 1000 mcg/day or placebo. The active group showed improvement in several nerve-conduction parameters, sudomotor function, pain score and quality of life, though cardiovascular autonomic reflex tests and one neuropathy exam score did not improve.

APPLICABILITY: Supports a nerve-related role only in metformin-treated diabetic patients with borderline B12 and existing neuropathy, using oral rather than IV dosing; it does not show benefit for nerve health in healthy people or via an IV drip.

SOURCE: Nutrients, 2021 (randomized double-blind placebo-controlled trial) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7912007/>

FACT-CHECK Both citations verified as real, correctly attributed studies (PMC pages returned 403 to the fetcher, but confirmed via PubMed/PMC search metadata and the article's own table URL). Kept both. Study 1 (Markun et al., Nutrients 2021, PMC8000524) was misrepresented: the original summary claimed it concluded "most fatigue complaints are not linked to B12 deficiency" and that fatigue benefit is "uncertain," but the review actually found no effect on cognition or mood and could not analyze fatigue at all (only one trial reported it). Rewrote the finding/summary/context to reflect the true negative cognition/mood result and the sparse fatigue evidence. Study 2 (Nutrients 2021, PMC7912007) was accurate; added detail that patients were metformin-treated with borderline B12 (<400 pmol/L) and that autonomic reflex tests did not improve, avoiding overstatement. evidenceLevel kept at "limited": one study is negative/inconclusive in non-deficient people and the other shows benefit only in a diabetic-neuropathy population with borderline B12, so evidence for B12 (especially IV) in healthy wellness clients remains limited. All language kept CPSO-safe ("may support"), no efficacy guarantees for healthy people or for the IV route.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Defined roles in metabolism and mood; the mood evidence is oral and modest, with no IV data.

1. High-dose oral B6 reduced self-reported anxiety

A randomized, double-blind, placebo-controlled trial in roughly 478 healthy young adults found that one month of high-dose oral vitamin B6 (100 mg/day, about 50x the RDA) may modestly reduce self-reported anxiety, with a non-significant trend toward lower depression, consistent with B6's role as a cofactor in GABA synthesis. This may support mood but does not test an IV route.

APPLICABILITY: Directly on mood but oral, high-dose, and short-term in healthy young adults; the anxiety effect was modest and the depression effect only a trend, and it does not evaluate IV administration, so it offers only indirect support for an IV wellness drip.

SOURCE: Human Psychopharmacology: Clinical and Experimental, 2022 (RCT) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC9787829/>

2. Magnesium plus B6 eased stress in low-magnesium adults

A post-hoc analysis of a randomized controlled trial reported that oral magnesium (300 mg/day) combined with vitamin B6 (30 mg/day) may improve perceived stress and quality of life more than magnesium alone in stressed, otherwise healthy adults with low magnesium levels. B6 was given only in combination, so its independent contribution to mood cannot be isolated.

APPLICABILITY: Oral combination product in people with low magnesium, not B6 alone and not IV; applicability to a B6 IV drip in already-replete healthy clients is weak.

SOURCE: Stress and Health, 2021 (post-hoc analysis of an RCT) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC9292249/>

3. B6 is a cofactor across metabolism; deficiency alters metabolic pathways

A controlled dietary study in 23 healthy adults showed that 28 days of marginal vitamin B6 restriction produced measurable changes in amino acid and related metabolic pathways, underscoring B6's role (as pyridoxal 5'-phosphate) as a cofactor in a wide range of enzymatic reactions. This reflects the effect of inducing deficiency, not a metabolic benefit of extra B6 in people who are already replete.

APPLICABILITY: Establishes that B6 is needed for normal metabolism, but the effect is about avoiding deficiency; it does not show that IV B6 enhances metabolism in people who already have adequate levels.

SOURCE: PLOS One, 2013 (controlled dietary restriction study) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3679127/>

FACT-CHECK All three citations verified as real, correctly-loading studies (confirmed via search metadata, journal records, and PubMed IDs, since PMC hosts are blocked by the session egress policy). Two journal attributions were wrong and were corrected: study 2 is Stress and Health (2021), not "Human Psychopharmacology, 2022"; study 3 is PLOS One (2013), not "Journal of Nutrition." Study 1's summary was softened because the depression benefit was only a non-significant trend (added exact dose/sample). Study 2 summary clarified the doses (300 mg Mg + 30 mg B6) and that B6 was only given in combination. Study 3 summary softened "disruptions in one-carbon metabolism" (the study actually found one-carbon metabolism relatively resilient) to "amino acid and related metabolic pathways." All summaries kept CPSO-safe "may support" language with no guarantees. evidenceLevel remains "limited": none tests an IV route or IV B6 in already-replete healthy people. 0 studies removed; 3 retained.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Essential for energy and nerves, but trials are in deficient or critically ill patients, not healthy people.

1. IV thiamine in critically ill patients: modest metabolic effects, no mortality benefit

Systematic review and meta-analysis of randomized controlled trials (35 studies, ~3,494 patients) of intravenous thiamine in critically ill adults. Thiamine may be associated with a modest reduction in lactate and shorter shock duration, but showed little to no difference in mortality and a slight increase in ICU length of stay; overall certainty of evidence was low.

APPLICABILITY: Directly uses the IV route, but the population is critically ill patients, not healthy people seeking an energy or wellness drip, so it does not show a benefit in healthy adults.

SOURCE: Systematic review & meta-analysis of RCTs, Clinical Nutrition / PubMed 2024 · <https://pubmed.ncbi.nlm.nih.gov/39307094/>

2. IV thiamine as metabolic resuscitator did not lower lactate overall in septic shock

Randomized, double-blind, placebo-controlled pilot trial (88 patients) of intravenous thiamine in septic shock. There was no difference in the primary outcome of lactate at 24 hours between thiamine and placebo, with a possible signal only in a pre-specified thiamine-deficient subgroup.

APPLICABILITY: Tests the energy-metabolism (lactate) rationale via IV thiamine, but in septic shock patients; the mostly null result and the healthy-vs-critically-ill gap mean it does not support IV thiamine for energy in well people.

SOURCE: Randomized double-blind placebo-controlled pilot RCT, Critical Care Medicine 2016 (PMC) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC4754670/>

3. Thiamine repletion may reverse deficiency-related neuropathy

Prospective therapeutic-challenge study of 41 adults from an endemic area (polished-rice diet) describing thiamine deficiency-related peripheral neuropathy that improved with thiamine repletion, consistent with thiamine's role in nervous-system function when deficiency is present.

APPLICABILITY: Suggests a nervous-system benefit only in people who were thiamine-deficient; it does not show that supplementing already-replete, healthy people supports the nervous system, and it is not an IV-drip study.

SOURCE: Prospective clinical study, European Journal of Neurology 2024 (PubMed) · <https://pubmed.ncbi.nlm.nih.gov/38018774/>

FACT-CHECK All 3 citations verified as real, correctly attributed studies whose findings match the claims. NCBI hosts were blocked by this session's egress policy for direct WebFetch (HTTP 403), so each was confirmed via web search, which returned the identical URLs and their content. Changes: (1) Study 1 summary tightened to add the no-mortality-benefit/low-certainty framing (added slight ICU-stay increase) — no overstatement found. (2) Study 3 was mislabeled as a 'case series' dated 2023; it is actually a prospective therapeutic-challenge study of 41 patients published in European Journal of Neurology 2024 — corrected design descriptor, year, and softened to 'may reverse'/'improved with' language. (3) Minor CPSO-safe wording throughout ('may be associated'). evidenceLevel kept at 'limited': every study is in deficient or critically ill populations, none support IV thiamine for energy/wellness in healthy, replete adults. No studies removed.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Strongly established for oral supplementation before and during early pregnancy; not an IV wellness claim.

1. Folic acid may reduce first-occurrence neural tube defects

This limited systematic review update for the U.S. Preventive Services Task Force found continued evidence that periconceptional ORAL folic acid supplementation may reduce the risk of neural tube defects in people who are pregnant or planning pregnancy, with no evidence of harm. The update drew mainly on observational cohort and case-control studies alongside earlier trial evidence; the route studied was oral and the population was people capable of pregnancy.

APPLICABILITY: Supports folic acid's role in healthy fetal neural development, but the evidence is for oral supplementation around pregnancy, not IV drips in healthy non-pregnant adults.

SOURCE: Agency for Healthcare Research and Quality / USPSTF limited systematic review update (NCBI Bookshelf) · <https://www.ncbi.nlm.nih.gov/books/NBK593614/>

2. Umbrella review supports NTD prevention

An umbrella review of 10 systematic reviews and meta-analyses (about 296,816 participants) reported that preconception folic acid supplementation was associated with a markedly lower risk of neural tube defects (folic-acid pooled effect approximately 0.23; roughly a 57% overall reduction). Dosing was mostly oral and preconceptional in women of reproductive age.

APPLICABILITY: Robust evidence for the developmental-support claim, but it reflects oral periconceptional dosing in women of reproductive age, not general IV wellness use in healthy people.

SOURCE: Umbrella review of systematic reviews and meta-analyses, Neuroepidemiology (Karger), 2024 · <https://pubmed.ncbi.nlm.nih.gov/39074459/>

3. Analysis shows reduced NTD recurrence

A 2007 analysis of randomized folic acid trials reported roughly a 69% reduction (intention-to-treat) and up to an 87% reduction (among women taking supplements before pregnancy) in recurrent neural tube defects with periconceptional oral supplementation, consistent with folate's role in cell division during early development.

APPLICABILITY: Strong evidence, but limited to oral supplementation in a high-risk pregnancy population; it does not evaluate IV folic acid or any benefit in healthy people seeking wellness infusions.

SOURCE: Grosse & Collins, Birth Defects Research Part A, 2007 (analysis of randomized trials) · <https://pubmed.ncbi.nlm.nih.gov/17990333/>

FACT-CHECK All 3 citations verified as real studies via web search (NCBI/PubMed are blocked for direct fetch by the session egress policy, so identity and figures were confirmed through search-result summaries). Kept all three. Changes: (1) NBK593614 - removed the unverifiable "RR ~0.38" and the "pooling RCTs and cohorts" characterization, since this limited update relied mainly on observational studies; softened "cuts" to "may reduce" for CPSO safety. (2) 39074459 - tightened the loose "~0.23-0.43" range to the confirmed folic-acid pooled effect (~0.23) and ~57% overall reduction, added sample size. (3) 17990333 - retained (69%/87% figures confirmed exactly); attributed authorship and softened phrasing. evidenceLevel remains "established" for oral periconceptional folic acid; every context line flags that the evidence is oral/pregnancy-specific and not for IV use in healthy adults.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: An essential vitamin, but hair and nail benefits beyond deficiency rest on small, uncontrolled studies.

1. Oral biotin improved brittle nails in an uncontrolled series

In a small retrospective, uncontrolled case series of patients with brittle nails, daily oral biotin was associated with subjective clinical improvement in roughly two-thirds (about 63%) of those who took it. There was no placebo control, the sample was small, and outcomes were self-reported, so this may at most suggest a possible benefit.

APPLICABILITY: Oral, not IV, and limited to people with pre-existing brittle nails; it provides no evidence that IV biotin benefits nails in healthy people. A frequently cited ~25% nail-thickness increase comes from a separate referenced study, not this one.

SOURCE: Cutis, 1993 (retrospective uncontrolled case series) · <https://pubmed.ncbi.nlm.nih.gov/8477615/>

2. Review finds weak evidence for biotin in nail disease

A narrative review of biotin for nail disease found the supporting literature to be small and largely uncontrolled, noting some improvement reported with oral biotin but concluding that larger controlled trials are needed before any benefit for nails can be considered established.

APPLICABILITY: Evaluates oral supplementation only; there is no IV data, and the review itself calls the overall evidence base weak and in need of controlled trials.

SOURCE: Journal of Dermatological Treatment, 2018 (narrative review) · <https://pubmed.ncbi.nlm.nih.gov/29057689/>

3. Biotin for hair loss lacks quality evidence outside deficiency

A systematic review of oral biotin for hair loss identified only three qualifying studies; the highest-quality, double-blind placebo-controlled study showed no difference between biotin and placebo for hair growth, and the review concluded that high-quality evidence does not demonstrate a clear hair benefit in people who are not biotin-deficient.

APPLICABILITY: Concerns oral biotin in generally non-deficient people; there is no evidence that IV biotin improves hair in healthy individuals.

SOURCE: Journal of Clinical and Aesthetic Dermatology (JCAD), 2024 (systematic review) · <https://pubmed.ncbi.nlm.nih.gov/39148962/>

FACT-CHECK All three citations are real and were verified via web search (direct PubMed/NCBI fetch was blocked by egress policy). None removed. Changes: (1) 8477615 — corrected the summary, which had misattributed a "~25% nail plate thickness" increase to this Cutis 1993 study; that figure actually comes from a separate referenced Swiss study (PMID 2273113). This study is a retrospective uncontrolled series showing ~63% subjective clinical improvement; summary/finding rewritten to reflect that and softened to CPSO-safe "may suggest" language. (2) 29057689 — corrected the journal from the incorrect "Skin Appendage Disorders" to the actual "Journal of Dermatological Treatment" (2018); summary tightened. (3) 39148962 — corrected the journal from the incorrect "Skin Appendage Disorders" to the actual "Journal of Clinical and Aesthetic Dermatology (JCAD)" (2024); confirmed it is a systematic review with the placebo-controlled study showing no benefit. evidenceLevel kept at "limited": all data are oral and low-quality/uncontrolled, and none of the studies address IV biotin or healthy individuals.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Skin evidence is real but oral or topical in higher-risk patients; there is no IV data in healthy people.

1. Oral nicotinamide cut new nonmelanoma skin cancers

In the ONTRAC phase 3 randomized controlled trial (386 immunocompetent adults with at least two nonmelanoma skin cancers in the prior 5 years), oral nicotinamide 500 mg twice daily for 12 months was associated with a 23% lower rate of new nonmelanoma skin cancers versus placebo, with no benefit seen after the drug was stopped. The route was oral, not IV.

APPLICABILITY: Relevant to skin, but it tested oral dosing in higher-risk patients with prior skin cancers, not IV infusion in healthy wellness clients; it does not show that an IV drip provides this benefit.

SOURCE: N Engl J Med, 2015 (Phase 3 RCT) · <https://pubmed.ncbi.nlm.nih.gov/26488693/>

2. Oral nicotinamide lowered transepidermal water loss

A randomized controlled trial (British Journal of Dermatology, 2016) reported that oral nicotinamide 500 mg twice daily produced a modest relative reduction in transepidermal water loss (about 5-6% on the forehead over 12 months), which the authors interpret as a possible skin-barrier benefit. Administration was oral, not intravenous.

APPLICABILITY: May support a small skin-barrier effect, but the finding is from oral supplementation and cannot be assumed to transfer to an IV route or to healthy people.

SOURCE: Br J Dermatol, 2016 (RCT) · <https://pubmed.ncbi.nlm.nih.gov/27062605/>

3. Review of niacinamide mechanisms and skin evidence

A 2021 review (Antioxidants) summarizes clinical evidence that niacinamide, mainly applied topically or taken orally, may help with skin aging and pigmentation via its role as an NAD⁺ precursor, while noting no confirmed specific molecular target. It is a narrative review, not a controlled trial.

APPLICABILITY: Useful background linking B3 to skin and NAD-dependent metabolism, but it is not IV-specific and provides no direct evidence that IV niacinamide boosts energy or skin outcomes in healthy individuals.

SOURCE: Review article, Antioxidants (MDPI), PMC 2021 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC8389214/>

FACT-CHECK All three citations were verified as real and accurately attributed (PubMed/PMC returned 403 to direct fetch, so each was confirmed via search of the indexed record). Study 1: ONTRAC NEJM 2015 phase 3 RCT confirmed (386 participants, 500 mg twice daily, 12 months, 23% reduction); softened "reduced by 23%" to "23% lower rate" and added that benefit did not persist after stopping. Study 2: confirmed as Chen et al., Br J Dermatol 2016 RCT; corrected the vague source to the actual journal/year and added the small effect size (5-6%) rather than an unqualified "improved barrier function." Study 3: confirmed as Boo 2021 review in Antioxidants (MDPI); relabeled the finding as a review and noted its own caveat about no confirmed molecular target. No studies removed. evidenceLevel downgraded from "emerging" to "limited": every source is oral or topical and in patients with skin-cancer history or aged skin — there is no IV-route and no healthy-population evidence, so the evidence for IV niacinamide as offered is limited. All summaries use "may support" / associational language and flag the oral-vs-IV gap for CPSO safety.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Skin and metabolic roles are established biochemically; human evidence is oral, topical, or mechanistic only.

1. Oral pantothenic acid may reduce acne lesion counts

A 12-week randomized, double-blind, placebo-controlled trial in healthy adults with mild-to-moderate facial acne reported that an oral pantothenic acid-based dietary supplement reduced total facial lesion count versus placebo, with a significant reduction in inflammatory lesions. Route was oral, not IV.

APPLICABILITY: Directly about skin, but it studied an oral supplement in people with active acne, not IV panthenol in healthy people, so it does not establish that an IV drip supports skin.

SOURCE: Dermatol Ther (Heidelb), 2014 (RCT) · <https://pubmed.ncbi.nlm.nih.gov/24831048/>

2. Topical panthenol mask may aid skin barrier recovery after laser

A double-blind randomized controlled study of a panthenol-enriched mask (also containing centella asiatica/madecassoside and bisabolol) after non-ablative fractional facial laser treatment reported significantly lower erythema and hyperpigmentation indices versus a saline dressing at days 3, 7, and 14. Route was topical.

APPLICABILITY: Supports a topical skin-barrier role, but the product was multi-ingredient and applied to the skin, not infused, and it says nothing about IV panthenol in healthy people.

SOURCE: J Cosmet Dermatol, 2025, PMC12232337 (RCT) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12232337/>

3. Vitamin B5 is a component of coenzyme A central to metabolism

This reference article summarizes that pantothenic acid is required to form coenzyme A, which is needed for metabolism of fatty acids, carbohydrates, proteins, and hormones, and notes that supplementation is primarily relevant to correcting deficiency, which is rare. It is a narrative reference, not an outcomes trial.

APPLICABILITY: Describes the well-established biochemical role of B5 in metabolism, but this is mechanism and deficiency background and does not show that IV B5 improves metabolism in already-replete healthy people.

SOURCE: StatPearls, NCBI Bookshelf (review/reference) · <https://www.ncbi.nlm.nih.gov/books/NBK563233/>

FACT-CHECK All 3 citations verified as real and correctly represented; none removed. NCBI and publisher (Springer/Wiley) hosts are blocked by this session's egress policy (403 at the gateway), so direct WebFetch of each URL failed; the README forbids routing around policy denials. I verified each via WebSearch, which returned the corresponding PubMed/publisher listings and abstract-level detail confirming study existence, design, route, and findings. Wording tightened for CPSO safety: study 1 summary now says total lesion count was reduced with inflammatory lesions significantly reduced (matches primary vs secondary outcomes) and specifies a healthy-adult population; study 2 source corrected to note the 2025 publication year and clarified the laser was non-ablative fractional. evidenceLevel kept at 'limited' because no citation involves IV panthenol in healthy people; the evidence base is oral, topical, or mechanistic only.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Important for muscle and nerve function, but trials for cramps and recovery are mixed to negative, including IV.

1. Little evidence magnesium prevents skeletal muscle cramps

This Cochrane systematic review pooled randomized trials of magnesium (oral and intravenous) for skeletal muscle cramps, largely in older adults but also including pregnancy-associated cramps. It concluded that magnesium is unlikely to provide clinically meaningful prevention of idiopathic cramps (moderate-certainty evidence), while evidence in pregnancy was low-certainty and conflicting.

APPLICABILITY: Directly relevant and high quality, but it examines cramp prevention (not general muscle 'recovery'), and its overall finding is that magnesium is unlikely to help. It does not support IV magnesium for muscle wellness in healthy people.

SOURCE: Cochrane Database of Systematic Reviews, 2020 (Garrison SR et al.; CD009402.pub3) · <https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.CD009402.pub3/full>

2. IV magnesium infusion did not reduce rest leg cramps

A double-blind, placebo-controlled randomized trial gave 46 community-dwelling older adults 5 consecutive days of intravenous magnesium sulfate (20 mmol/5 g) versus placebo. Intravenous magnesium did not reduce the frequency of rest (nocturnal) leg cramps, regardless of how much infused magnesium was retained.

APPLICABILITY: One of the few trials using the IV route specifically, but it was in older adults with cramps and found no benefit, so it does not support an IV muscle-recovery claim in healthy individuals. (Note: the journal was corrected from an earlier mislabel of 'Medical Journal of Australia'.)

SOURCE: The Journals of Gerontology: Series A, 2011 (PMID 21289017) · <https://pubmed.ncbi.nlm.nih.gov/21289017/>

3. IV magnesium may improve post-surgical quality of recovery

A meta-analysis of 7 randomized controlled trials (622 patients) found that perioperative intravenous magnesium sulfate may improve patients' subjective postoperative quality-of-recovery scores on day 1 after surgery under general anesthesia, with associated reductions in opioid use and postoperative nausea/vomiting.

APPLICABILITY: Uses the IV route and shows a recovery benefit, but the population is surgical patients under anesthesia; it may not generalize to muscle recovery in healthy people receiving elective IV wellness drips.

SOURCE: Nutrients, 2024 (meta-analysis of RCTs; PMID 39064818; PMC11280250) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11280250/>

FACT-CHECK All 3 citations verified as real studies whose findings match their summaries; none removed. Direct WebFetch to pubmed/ncbi/cochranelibrary was blocked by the session's egress policy (403 at proxy gateway), so verification was done via WebSearch, which independently confirmed each title, design, population, and result. Changes: (1) Study 2's source was corrected from 'Medical Journal of Australia' to 'The Journals of Gerontology: Series A, 2011' — PMID 21289017 maps to that journal, not MJA. (2) Study 3's source tightened to name the journal (Nutrients, 2024; 7 RCTs/622 patients) and PMID 39064818. (3) Minor CPSO-safe wording tweaks to summaries (kept 'unlikely to help' and 'may improve' hedging). evidenceLevel retained as 'limited': the two cramp trials show no benefit and the only positive trial is in surgical patients, so evidence for IV magnesium for muscle recovery/wellness in healthy people is limited-to-absent.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: May shorten colds and aid wound healing (oral or topical); no IV evidence in healthy, zinc-replete people.

1. **Oral zinc may shorten common cold duration**

A 2024 Cochrane systematic review of randomized trials in mostly healthy adults and children found that oral (lozenge or syrup) zinc taken as treatment may reduce the duration of common cold symptoms, but confidence in this finding was low and effects on prevention were uncertain. All included trials used oral, not intravenous, zinc.

APPLICABILITY: Points to a possible immune-related role of oral zinc in the general population; it does not test intravenous zinc, and it does not show benefit in already zinc-replete healthy people.

SOURCE: Cochrane Database of Systematic Reviews, 2024 (CD014914.pub2) · <https://www.cochranelibrary.com/cdsr/doi/10.1002/14651858.cd014914.pub2/full>

2. **Zinc may support wound and ulcer healing**

A 2025 systematic review and meta-analysis of five clinical trials found that zinc treatment may be associated with improved ulcer healing at the final endpoint, with GRADE-rated moderate-quality evidence tempered by study limitations. Trials used oral or topical zinc in patients with existing wounds, not intravenous infusion.

APPLICABILITY: Relevant to skin-repair claims but studied in patients with existing wounds using oral or topical routes; there are no data here on intravenous zinc or on skin repair in healthy people.

SOURCE: Systematic review and meta-analysis, BMJ Nutrition, Prevention & Health, 2025 (PMID 40771531) · <https://pubmed.ncbi.nlm.nih.gov/40771531/>

FACT-CHECK Verified all 3 citations. Kept 2, removed 1. (1) Cochrane CD014914 (2024) confirmed real; softened certainty from "low to moderate" to "low" since the duration finding is low-certainty, and corrected URL to the reviewed /full page. (2) PMID 40771531 confirmed real (BMJ Nutr Prev Health 2025, 5 trials, moderate GRADE); kept, source publisher added. (3) REMOVED PMID 28395131 (Momen-Heravi/Asemi 2017 diabetic foot ulcer RCT): the study is real but carries a formal Expression of Concern from the journal (Wound Repair and Regeneration 2024, doi 10.1111/wrr.12537); senior author Z. Asemi has numerous retractions/expressions of concern for data-integrity issues, making it unsafe to cite. Downgraded evidenceLevel from "emerging" to "limited": surviving evidence is oral/topical only, low-to-moderate certainty, with no data on IV administration or on healthy zinc-replete individuals. Note: NCBI/Cochrane/Wiley were blocked by the session egress proxy for direct fetch (403 policy denials, not retried per proxy rules); existence and details were confirmed via web search of the primary source listings.

☐ Approve as written ☐ Revise (notes below)

Notes: _____

PUBLISHES AS: Well-defined for bone, but supplement trials show small, oral, vitamin-D-dependent effects, not IV.

1. Oral calcium produces only small, non-progressive gains in bone density

Systematic review and meta-analysis of randomized controlled trials in adults over 50 found that increasing calcium intake via diet or oral supplements raised bone mineral density by roughly 0.6-1.8% over 1-5 years, an effect the authors judged unlikely to meaningfully reduce fracture risk. All data were oral intake; no IV route was studied.

APPLICABILITY: Directly relevant to the bone claim but tests oral calcium in older adults; it does not support an IV drip in healthy people, and even for oral use the bone-density effect is small and does not progress over time.

SOURCE: BMJ systematic review and meta-analysis, 2015 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC4784773/>

2. Oral calcium plus vitamin D may modestly lower fracture risk in older adults

Updated National Osteoporosis Foundation meta-analysis of randomized controlled trials reported that oral calcium combined with vitamin D was associated with roughly a 15% reduction in total fractures and a 30% reduction in hip fractures, mainly in middle-aged to older adults. The benefit was for the oral calcium-plus-vitamin-D combination, not calcium alone and not by IV.

APPLICABILITY: May support bone health only for oral calcium co-administered with vitamin D in older or at-risk people; it cannot be read as evidence that IV calcium benefits bones or muscle in healthy clients.

SOURCE: Osteoporosis International meta-analysis (NOF), 2016 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC4715837/>

3. Muscle/strength benefits in trials track vitamin D repletion, not calcium alone

Randomized controlled trial in vitamin D-deficient adults found that six months of oral cholecalciferol plus calcium was associated with improved skeletal muscle strength and physical performance. Because both nutrients were given together, calcium could not be isolated as the active agent, and the authors framed the improvement around correcting vitamin D deficiency.

APPLICABILITY: Weak support for the muscle claim: benefit tracks vitamin D repletion in deficient people, uses the oral route, and does not show that IV calcium improves muscle function in healthy individuals.

SOURCE: Clinical Endocrinology (Oxford) RCT, 2010 · <https://pubmed.ncbi.nlm.nih.gov/20455886/>

FACT-CHECK All 3 studies are real and verified (NCBI direct fetch was blocked by egress policy, so each was confirmed via web search against its title, journal, year, and results). All 3 kept, with corrections. Study 1: fixed "randomized and observational studies" to "randomized controlled trials" and corrected the BMD gain from "1-2%" to the actual "0.6-1.8%"; trimmed source to "BMJ." Study 2: accurate, kept essentially unchanged (minor wording to "may support"). Study 3: corrected a citation error, the journal is Clinical Endocrinology (Oxford), not the Journal of Clinical Endocrinology & Metabolism (different journal); also softened the summary since calcium and vitamin D were co-administered and cannot be separated, and noted the population was young deficient adults, not older. evidenceLevel kept at "limited" - all evidence is oral-route, none tests IV calcium, and benefits depend on vitamin D co-administration, so extrapolation to an IV drip in healthy people is not supported. Language kept CPSO-safe ("may support," "associated with," no guarantees).

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: The antioxidant role is recognised, but skin-tone evidence is low-quality oral/topical and IV use is not supported.

1. Oral glutathione may modestly lighten sun-exposed skin in some people

A 2020 review appraised three RCTs of systemic (oral) glutathione in adults: one trial did not support a skin-whitening effect, while two showed a reduced skin melanin index only in certain body areas or age groups. Oral preparations were reported as well tolerated but parenteral (injected) forms were not, and overall evidence quality was low. No RCT evidence supported the intravenous route.

APPLICABILITY: Relevant to the antioxidant/skin-tone claim, but it appraises oral (systemic) glutathione in mostly healthy adults and explicitly does not support the IV drip our clinic offers.

SOURCE: Dermatology Research and Practice / review, PMC (2020) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC7196133/>

2. Combined oral + topical glutathione lowered melanin index vs placebo

A double-blind randomized controlled trial in healthy adults found that combined oral (500 mg/day) plus topical glutathione produced a significantly lower melanin index than placebo, suggesting the combination may support a lighter, more even skin tone. Intravenous administration was not tested.

APPLICABILITY: An actual RCT for the skin-tone benefit, but in healthy volunteers using oral plus topical dosing, so it does not demonstrate that an IV glutathione drip produces the same effect.

SOURCE: Double-blind RCT, PubMed (2021) · <https://pubmed.ncbi.nlm.nih.gov/33871071/>

3. IV glutathione for skin lightening lacks efficacy and adequate safety data

A 2016 commentary titled 'Intravenous glutathione for skin lightening: inadequate safety data' noted there are no published studies of intravenous glutathione for skin lightening and that long-term safety data are lacking. It called for regulatory assessment of systemic/IV glutathione used cosmetically to protect consumers from potential harms of IV infusions.

APPLICABILITY: Highly relevant honesty check: it specifically addresses the IV route used in wellness settings and finds the evidence inadequate rather than supportive.

SOURCE: Commentary/letter, PubMed (2016) · <https://pubmed.ncbi.nlm.nih.gov/27499402/>

FACT-CHECK All 3 citations are real and verified (NCBI was blocked by egress policy, so confirmed via multiple independent sources). Corrections: (1) Study 1's journal was wrong — it is Dermatology Research and Practice (Sitohang & Ninditya, 2020), not Journal of Dermatological Treatment; summary trimmed of the unsupported "topical" framing since the review covers systemic/oral glutathione, and reworded to reflect that one of three RCTs opposed the effect. (2) Study 3's source/title corrected to the 2016 commentary "Intravenous glutathione for skin lightening: inadequate safety data," and the overstated "prompted regulatory warnings" softened to "called for regulatory assessment," which is what the article actually says. (3) Study 2 verified accurate and kept unchanged. All summaries use CPSO-safe "may support" language. evidenceLevel remains "limited" — appropriate given low study quality for oral/topical and no efficacy evidence for the IV route the clinic offers.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Central to cellular energy in biology, but human clinical benefits so far are limited and inconsistent.

1. IV NAD+ RCT in ischemic heart failure patients

A randomized, placebo-controlled trial gave intravenous NAD+ to patients with heart failure from ischemic cardiomyopathy. The NAD+ group showed a greater improvement in cardiac function (left-ventricular ejection fraction) at 1 month, along with changes in oxidative-stress and sirtuin markers, though some secondary endpoints (such as NT-proBNP) did not reach statistical significance. This suggests IV NAD+ may support cardiac and cellular pathways in a sick cardiac population, and does not demonstrate an energy benefit in healthy adults.

APPLICABILITY: Directly uses the IV route, but the population is heart-failure patients and some endpoints were not statistically significant, so it does not show an energy benefit in otherwise healthy people getting a wellness drip.

SOURCE: Am J Cardiovasc Drugs (randomized placebo-controlled trial), 2025 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC12779688/>

2. Oral NAD+ precursor (NR) raises NAD+ but limited functional effect

A 2x6-week randomized, double-blind, placebo-controlled crossover trial found the oral NAD+ precursor nicotinamide riboside (500 mg twice daily) was well tolerated and raised NAD+ levels in healthy middle-aged and older adults. Broader reviews note precursors reliably raise NAD+ yet show limited or inconsistent effect on energy expenditure or exercise capacity.

APPLICABILITY: This is an ORAL precursor, not IV NAD+; raising blood NAD+ is documented but a clear energy or performance benefit is not, so it cannot prove an IV drip 'works'.

SOURCE: Nature Communications (randomized double-blind placebo-controlled crossover), 2018 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC5876407/>

3. Review: limited clinical efficacy of NAD+ boosting in humans

A review of NAD+ precursor supplementation in human ageing concludes that despite strong preclinical rationale, precursors reliably raise NAD+ levels but human trials have shown limited and inconsistent clinical efficacy across outcomes such as insulin sensitivity, energy expenditure, and exercise capacity.

APPLICABILITY: Honest overview showing NAD+'s energy-metabolism role is established biochemically, but proven clinical energy benefits in people (oral or IV) remain weak and inconclusive.

SOURCE: Nature Metabolism review, 2025 · <https://pubmed.ncbi.nlm.nih.gov/41083806/>

FACT-CHECK All three citations verified as real via web search (PMC/PubMed 403 on direct fetch). Study 1 = Am J Cardiovasc Drugs 2025 IV NAD+ ischemic-cardiomyopathy RCT; its summary was corrected because the headline finding was improved cardiac function (LVEF), not just oxidative-stress markers, and some secondary endpoints were non-significant. Study 2 = NR crossover trial (Nature Communications 2018), source corrected to Nature Communications; summary accurate. Study 3 = Nature Metabolism 2025 review (title 'NAD+ precursor supplementation in human ageing: clinical evidence and challenges'); source corrected from 'PubMed review article' to Nature Metabolism. No studies removed. Summaries softened to CPSO-safe 'may support'/'limited and inconsistent' language. evidenceLevel kept at 'limited' since the only IV human data is in sick cardiac patients with partly non-significant endpoints and broader human evidence is weak.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Oral BCAAs may ease exercise soreness under specific conditions; an IV or at-rest wellness benefit is unproven.

1. BCAA supplementation may reduce muscle soreness and damage markers

Systematic review and meta-analysis of RCTs in mostly healthy, active adults found oral branched-chain amino acid supplementation may attenuate post-exercise muscle soreness (DOMS) and some muscle-damage biomarkers such as creatine kinase, though the size and timing of effects varied across studies.

APPLICABILITY: Evidence is for ORAL BCAAs, not IV amino acid drips; participants were healthy exercisers around exercise-induced muscle damage. The review does not test intravenous administration or any general 'recovery' benefit at rest.

SOURCE: Systematic review and meta-analysis with meta-regression, Sports Medicine - Open, 2024 (PMC) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11021390/>

2. BCAA benefit on muscle damage is conditional, not guaranteed

Systematic review concluded oral BCAA supplementation may help alleviate exercise-induced muscle damage only under specific conditions: damage of low-to-moderate extent, a high daily intake (over ~200 mg/kg/day) sustained for a prolonged period (over ~10 days), and dosing before the damaging exercise.

APPLICABILITY: Oral supplementation over days-to-weeks, not a single IV infusion; underscores that any benefit is narrow and dependent on dose and timing, so IV wellness dosing cannot be assumed equivalent.

SOURCE: Systematic review, Nutrients 2017 (PMC/PubMed) · <https://pmc.ncbi.nlm.nih.gov/articles/PMC5691664/>

3. Free amino acids raise blood levels but did not increase muscle protein synthesis more than protein

Double-blind RCT in 24 healthy young adults found ingesting free amino acids produced faster absorption and greater postprandial plasma amino acid availability than an equivalent amount of intact milk protein, yet muscle protein synthesis rates were no higher than with intact protein.

APPLICABILITY: Tests oral free amino acids, not IV; relevant because it shows that raising blood amino acid levels (the main thing an IV drip achieves) did not translate into greater muscle building in healthy young adults.

SOURCE: Double-blind randomized trial, J Nutr 2022 (PMC) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8754581/>

FACT-CHECK All three citations verified as real and accurately represented; none removed. PMC direct fetch returned 403, so each was confirmed via web search against PMC/PubMed/journal listings. Titles, journals, years, populations, oral route, and stated conclusions all match. Minor edits only: added the specific dose/duration thresholds (>200 mg/kg/day, >10 days) already implied in study 2, specified n=24 and the intact-milk-protein comparator in study 3, and tightened study 1's DOMS/CK wording. No summaries were overstated; study 1 if anything understates its own significant effects, which is CPSO-safe. evidenceLevel kept at 'limited' because every study tests ORAL amino acids in healthy exercisers around exercise; none tests IV administration or a general at-rest wellness benefit, so the drip application remains indirect.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: A known role in fat metabolism, but weight and performance effects are modest, oral, and inconsistent.

1. Oral L-carnitine modestly reduced body weight and fat mass

A 2020 systematic review and meta-analysis of 37 randomized controlled trials (2,292 participants, mostly adults with overweight or obesity) found oral L-carnitine supplementation was associated with modest reductions in body weight (about 1.2 kg), BMI, and fat mass. All trials used oral dosing, not IV, and the effect was most evident in people with overweight or obesity.

APPLICABILITY: Oral, not IV, and conducted mainly in people with overweight/obesity; it does not establish that an IV drip enhances fat-for-energy use in otherwise healthy people.

SOURCE: Clinical Nutrition ESPEN, 2020 systematic review and meta-analysis of 37 RCTs · <https://pubmed.ncbi.nlm.nih.gov/32359762/>

2. L-carnitine may reduce waist circumference and systolic blood pressure

A systematic review and meta-analysis of RCTs (oral L-carnitine, roughly 0.75-3 g/day for 8-24 weeks) found supplementation was associated with modest reductions in waist circumference and systolic blood pressure. It reported no significant effect on fasting blood glucose, triglycerides, or HDL cholesterol. Populations were largely those with metabolic risk, using oral supplementation.

APPLICABILITY: Evidence is oral and in metabolic-syndrome populations, and benefits were limited to waist circumference and systolic blood pressure rather than glucose or lipids; relevance to IV wellness in otherwise healthy people is indirect and unproven.

SOURCE: Systematic review and meta-analysis of RCTs, PMC 2020 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC7551203/>

3. Exercise fat-oxidation benefit is proposed but inconsistent

A 2021 systematic review of acute and chronic oral L-carnitine supplementation for exercise performance notes that, because carnitine availability can limit fatty-acid oxidation during exercise, supplementation has been proposed to support fat use for energy. Prolonged high-dose oral dosing may increase muscle carnitine and fat oxidation during low-intensity exercise, but results across studies are mixed and intensity-dependent.

APPLICABILITY: Oral supplementation in exercising adults with inconsistent findings; provides no direct evidence for IV administration or for healthy people at rest.

SOURCE: Systematic review, Nutrients (MDPI) / PMC 2021 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC8704793/>

FACT-CHECK All three citations point to real, verifiable reviews and were kept (0 removed). NCBI hosts were blocked by the session egress policy, so each was confirmed via the search index (title/abstract metadata) rather than direct fetch. Changes: (1) Study 1 source corrected from the inaccurate "Obesity Reviews / Pharmacol Res" to the actual journal, Clinical Nutrition ESPEN 2020; summary lightly tightened. (2) Study 2 was misrepresented — the citation claimed improvements in "fasting glucose and lipid-related biomarkers," but PMC7551203 found significant reductions only in waist circumference and systolic blood pressure and NO significant effect on fasting glucose, triglycerides, or HDL. Finding, summary, and context rewritten to match the actual results and the dose range corrected to 0.75-3 g/day. (3) Study 3 accurately represented; only minor wording/source cleanup. evidenceLevel kept at "limited": all evidence is oral and in overweight/obese or metabolic-risk populations, with no direct IV or healthy-person data.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Individual components show oral effects on blood lipids in patients; injectable MIC for fat loss is unproven.

1. Oral inositol improved lipid profiles in metabolic patients

A 2018 systematic review and meta-analysis of 14 RCTs found that oral inositol supplementation significantly lowered triglycerides, total cholesterol and LDL-cholesterol in patients with metabolic diseases (no effect on HDL). It measured blood lipids, not body-fat loss.

APPLICABILITY: Only one of three MIC components (inositol), taken orally in people with existing metabolic disease; it may support healthier blood lipids in that group but does not test injectable MIC or fat loss in healthy clients.

SOURCE: Lipids in Health and Disease / PMC, 2018 (systematic review & meta-analysis of RCTs) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5968598/>

2. Oral choline aided liver fat markers in NAFLD patients

A 2025 randomized controlled trial gave oral phosphatidylcholine (2400 mg/day for 12 weeks) to patients with non-alcoholic fatty liver disease and reported favourable changes in hepatic steatosis, liver stiffness, oxidative stress markers and triglycerides versus controls. The population was a liver-disease group, not healthy adults.

APPLICABILITY: Choline (one MIC component) given orally to NAFLD patients; relevant to choline's role in liver fat handling, but it does not establish that an IV MIC drip reduces body fat in healthy people.

SOURCE: Therapeutic Advances in Chronic Disease / PubMed, 2025 (randomized controlled trial) · <https://pubmed.ncbi.nlm.nih.gov/40838115/>

3. Oral inositol plus alpha-lipoic acid improved metabolic markers

A 6-month randomized placebo-controlled trial in 155 postmenopausal women with metabolic syndrome found that oral inositol combined with alpha-lipoic acid was associated with a greater than 20% reduction in insulin resistance (HOMA-IR) in about two-thirds of the supplement group and lower triglycerides in a subset, compared with placebo.

APPLICABILITY: Tests inositol combined with a different compound (alpha-lipoic acid), taken orally in women with metabolic syndrome; it may support metabolic markers but is not the injectable MIC blend and does not measure body-fat loss.

SOURCE: Trials / PMC, 2013 (randomized placebo-controlled trial) · <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3765513/>

FACT-CHECK All 3 citations verified as real studies that load and genuinely support their findings; none removed. NCBI/PubMed were unreachable via direct fetch (proxy gateway 403), so each was confirmed via web search of title, authors, journal, sample and results. Corrections made: (1) Study 1 journal was wrong — it is 'Lipids in Health and Disease' (2018), not 'Int J Endocrinol'; added the null HDL result. (2) Study 2 year corrected 2024 to 2025, journal identified as Therapeutic Advances in Chronic Disease, and dose/route (oral phosphatidylcholine 2400 mg/day, 12 wk) added for accuracy. (3) Study 3 journal was wrong — it is 'Trials' (2013), not 'Gynecol Endocrinol'; summary tightened to the actual reported figures (HOMA-IR fell >20% in ~two-thirds; triglycerides down in a subset). All summaries kept CPSO-safe ('may support', no guarantees) and contexts retain the honest caveats that these test single MIC components, mostly orally, in disease populations — not injectable MIC or fat loss in healthy people. evidenceLevel left at 'limited', which remains appropriate.

☐ Approve as written ☐ Revise (notes below)

Notes:

PUBLISHES AS: Strong clinical evidence for treating diagnosed iron deficiency; it is not a wellness drip for replete people.

1. IV iron outperforms oral iron across multiple conditions

Systematic review and meta-analysis of 31 studies in patients with iron deficiency anemia found IV iron produced significantly greater hemoglobin increases than oral iron in CKD, IBD, cancer-related and general IDA, with fewer GI adverse events; no significant difference was seen for postoperative anemia or restless legs syndrome. Applies to diagnosed deficiency, not general supplementation.

APPLICABILITY: Population is people with diagnosed iron deficiency anemia, which matches the 'supplements are not enough' framing; it does not speak to IV iron in healthy people without deficiency.

SOURCE: Systematic review and meta-analysis (PMC), 2024 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC12635910/>

2. IV iron sucrose raised hemoglobin more than oral iron in pregnancy

RCT in pregnant women (14-34 weeks) with iron deficiency anemia (Hb 70-110 g/L, ferritin <15) found IV iron sucrose produced a larger hemoglobin rise at 4 weeks than oral ferrous fumarate (22 vs 12 g/L, $p < 0.0001$), with a greater rise in serum ferritin. May support IV iron use when oral iron is inadequate or not tolerated.

APPLICABILITY: Specific to anemic pregnant patients with confirmed deficiency; results should not be generalized to non-deficient adults seeking a wellness drip.

SOURCE: Randomized controlled trial, BMC Pregnancy and Childbirth (PMC), 2017 · <https://pmc.ncbi.nlm.nih.gov/articles/PMC5422878/>

3. IV iron gives modestly greater hemoglobin gain than oral iron in CKD

Updated systematic review and meta-analysis (Shepselovich et al.) in chronic kidney disease found IV iron achieved a modestly higher hemoglobin than oral iron (weighted mean difference ~0.31 g/dL in non-dialysis CKD and ~0.83 g/dL in dialysis patients), with no significant difference in mortality or serious adverse events. May support IV use where oral iron is insufficient in iron-deficient CKD.

APPLICABILITY: Studied CKD patients with deficiency; establishes benefit only where a true iron deficit exists, not for otherwise iron-replete individuals.

SOURCE: Systematic review and meta-analysis, American Journal of Kidney Diseases (AJKD)/PubMed, 2016 · <https://pubmed.ncbi.nlm.nih.gov/27321965/>

FACT-CHECK All 3 citations are real and correctly matched to their URLs (verified via web search; NCBI direct-fetch is blocked by this session's egress proxy, not a study failure). Kept all 3. Study 1 (PMC12635910): confirmed — 31 studies, IV iron superior in CKD/IBD/cancer/IDA with fewer GI effects; added the honest null results (no benefit for postoperative anemia or restless legs) so it isn't overstated. Study 2 (PMC5422878): confirmed 22 vs 12 g/L; corrected source to name the journal (BMC Pregnancy and Childbirth), removed the unverifiable ">20 g/L improvement" sub-claim, and added the actual inclusion criteria. Study 3 (PMID 27321965): TWO fixes — (a) journal was wrong, it is American Journal of Kidney Diseases (Shepselovich et al.), NOT Kidney International; (b) the "~0.8 g/dL" figure was overstated/cherry-picked — that is the dialysis subgroup; the non-dialysis WMD is only ~0.31 g/dL, so I reported both and added the null mortality/adverse-event finding. evidenceLevel left as "established": IV-vs-oral iron in diagnosed deficiency is well supported by multiple meta-analyses/RCTs. Caveat for the physician: none of these studies involve healthy, iron-replete people, so they cannot support IV iron as a general wellness drip.

☐ Approve as written ☐ Revise (notes below)

Notes:

OVERALL SIGN-OFF

☐ I approve the evidence framing above for publication. ☐ I approve with the revisions noted per ingredient.

Dr. Johnny Nguyen — Medical Director (signature)

Date

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